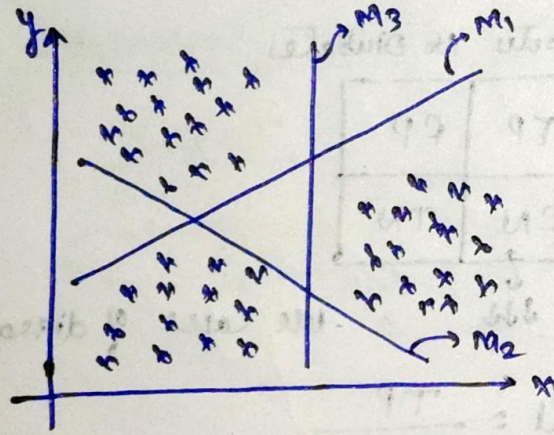


* Logistic Regression (One Versus Rest):

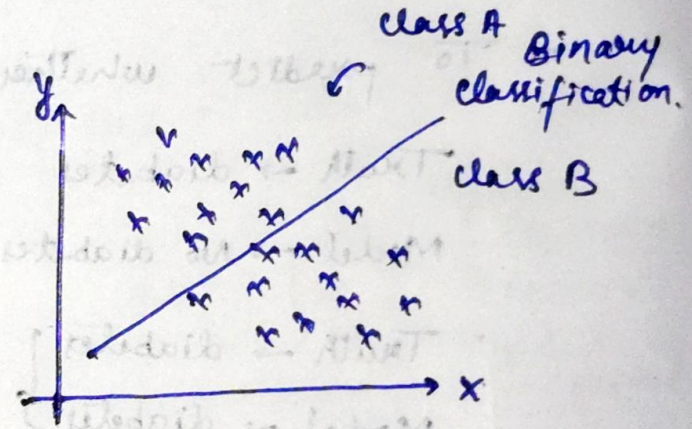


Multiclass classification
{ Logistic Regression }

$M_1 \rightarrow$ Binary classification

$M_2 \rightarrow$ Binary classification

$M_3 \rightarrow$ Binary classification



One Versus One

One Versus Rest (OVR) $\rightarrow$ Logistic Regression

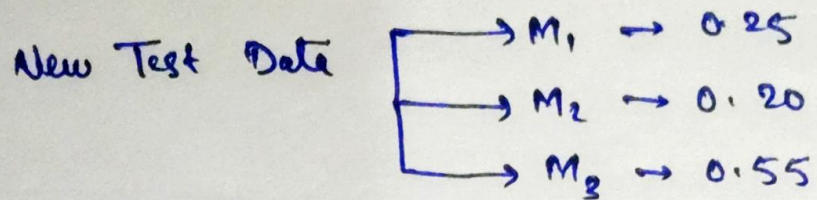
IP $\rightarrow$	f_1	f_2	f_3	O/P $\rightarrow 0_1, 0_2, 0_3$	0_1	0_2	0_3
-	-	-	-	0_1	1	0	0
-	-	-	-	0_2	0	1	0
-	-	-	-	0_3	0	0	1
-	-	-	-	0_1	1	0	0
-	-	-	-	0_3	0	0	1
-	-	-	-	0_2	0	1	0

Models

$$M_1 \leftarrow \text{inp } \{f_1, f_2, f_3\} \quad \text{oup}[\{0, 1\}]$$

$$M_2 \leftarrow \text{inp } \{f_1, f_2, f_3\} \quad \text{oup}[\{0, 2\}]$$

$$M_3 \leftarrow \text{inp } \{f_1, f_2, f_3\} \quad \text{oup}[\{0, 3\}]$$



$$0.55 \rightarrow \text{oup} \Rightarrow 0_3 \rightarrow \text{category 3}$$

$$[0.25, 0.20, \boxed{0.55}]$$

